Assuming that the discretization schemes preserve the ergodic property of Langevin dynamics, taking the expectation of these equations and considering the limit as n goes to infinity allows us to compute the invariant average of the schemes, and thus to evaluate the perfect sampling bias introduced by the discretization.

7.4.1 Invariant Averages for Schemes Using the A, B and O Splitting

The analysis of schemes using the splitting strategy coded with the A, B and O pieces is particularly simple, as we use a new random number for each stochastic update meaning that the random numbers and state variables are not correlated between steps. Hence we have

$$\mathbb{E}(\mu_{n,i} q_n) = \mathbb{E}(\mu_{n,i} p_n) = 0,$$

for any i and n. Thus, taking expectations of (7.8)–(7.10) for such schemes, and then considering the limit $n \to \infty$ we obtain a closed, linear system of equations

$$\begin{aligned}
\langle q^2 \rangle_h &= \psi_{11}^2 \langle q^2 \rangle_h + \psi_{12}^2 \langle p^2 \rangle_h + 2\psi_{11}\psi_{12} \langle qp \rangle_h + \bar{\mu}_{11}, \\
\langle p^2 \rangle_h &= \psi_{21}^2 \langle q^2 \rangle_h + \psi_{22}^2 \langle p^2 \rangle_h + 2\psi_{21}\psi_{22} \langle qp \rangle_h + \bar{\mu}_{22}, \\
\langle qp \rangle_h &= \psi_{11}\psi_{21} \langle q^2 \rangle_h + \psi_{12}\psi_{22} \langle p^2 \rangle_h + (\psi_{11}\psi_{22} + \psi_{12}\psi_{21}) \langle qp \rangle_h + \bar{\mu}_{12},
\end{aligned} \tag{7.11}$$

where we have constants

$$\bar{\mu}_{11} = \mathbb{E}(\mu_{n,1}^2), \quad \bar{\mu}_{12} = \mathbb{E}(\mu_{n,1}\mu_{n,2}), \quad \bar{\mu}_{22} = \mathbb{E}(\mu_{n,2}^2).$$

We can solve this linear system to find the invariant averages computed by the chosen discretization scheme for the harmonic oscillator with Hamiltonian (7.6).

Example 7.1 Consider the scheme coded ⟦ABO⟧, where

$$\begin{aligned}
&\llbracket \text{ABO} \rrbracket \\
&\boldsymbol{q}_{n+1} = \boldsymbol{q}_n + h\boldsymbol{M}^{-1}\boldsymbol{p}_n \\
&\boldsymbol{p}_{n+1/2} = \boldsymbol{p}_n - h\nabla U(\boldsymbol{q}_{n+1}) \\
&\boldsymbol{p}_{n+1} = e^{-h\gamma}\boldsymbol{p}_{n+1/2} + \zeta \boldsymbol{M}^{1/2}\mathbf{R}_n
\end{aligned}$$

with $\zeta = \left[\mathrm{k_B T}\left(1 - e^{-2\gamma h}\right)\right]^{1/2}$, and $\mathbf{R}_n$ a vector of unit normal random numbers. We can write this update scheme succinctly as

$$\begin{aligned}\boldsymbol{q}_{n+1} &= \boldsymbol{q}_n + h\boldsymbol{M}^{-1}\boldsymbol{p}_n,\\ \boldsymbol{p}_{n+1} &= e^{-h\gamma}\boldsymbol{p}_n - he^{-h\gamma}\nabla U(\boldsymbol{q}_n + h\boldsymbol{M}^{-1}\boldsymbol{p}_n) + \zeta\boldsymbol{M}^{1/2}\mathbf{R}_n.\end{aligned}$$

In the case of the one-dimensional harmonic oscillator, we have $U'(q) = \Omega^2 q$, yielding the compact form of one iteration as

$$\begin{bmatrix} q_{n+1} \\ p_{n+1} \end{bmatrix} = \begin{bmatrix} 1 & h/m \\ -\Omega^2 h e^{-h\gamma} & (1 - \Omega^2 h^2/m)e^{-h\gamma} \end{bmatrix} \begin{bmatrix} q_n \\ p_n \end{bmatrix} + \begin{bmatrix} 0 \\ \zeta m^{1/2}\mathrm{R}_n \end{bmatrix}. \tag{7.12}$$

Taking expectations of (7.12) and then taking the limit as $n \to \infty$, we see immediately that

$$\langle q \rangle_h^{[\![\mathrm{ABO}]\!]} = \langle p \rangle_h^{[\![\mathrm{ABO}]\!]} = 0,$$

as $\mathbb{E}(\mathrm{R}_n) = 0$. Plugging the values for ψ_{ij} into (7.11), with

$$\bar{\mu}_{11} = \bar{\mu}_{12} = 0, \quad \bar{\mu}_{22} = m\zeta^2,$$

we can solve to find the exact value of the method's stationary averages. For example, the relative increase in the variance of position is computed for the $[\![\mathrm{ABO}]\!]$ scheme as

$$\begin{aligned}\frac{\langle q^2 \rangle_h^{[\![\mathrm{ABO}]\!]}}{\langle q^2 \rangle} &= \frac{e^{h\gamma} m(1 + e^{-h\gamma})^2}{2m(1 + e^{-h\gamma}) - h^2\Omega^2 e^{-h\gamma}}\\ &= 1 + \gamma h/2 + (\Omega^2 + \gamma^2 m)h^2/(4m) + \mathcal{O}(h^3).\end{aligned}$$

Hence we can see that the $[\![\mathrm{ABO}]\!]$ scheme gives a first-order error in the variance of q. For significantly small friction (relative to h), the leading order term will be dominated by the term at order h^2, potentially giving us greater accuracy than we would expect from a first order scheme.

However, we see that the error grows exponentially as γ gets large. We can explain this by returning to the definition of the scheme, which has an O step (stochastic fluctuation) immediately preceding the only A step (drift), wiping out all force information for infinite friction. In practice, playing with the friction coefficient can affect the rate at which averages converge towards the invariant average.

From the work in Chap. 3, we may expect that performing the integration steps symmetrically (by using a Strang splitting) could provide desirable higher order behavior. Computing the coefficients in (7.11) for the each of the symmetric